

ANALYTICAL REPORT

Eurofins TestAmerica, Michigan
10448 Citation Drive
Suite 200
Brighton, MI 48116
Tel: (810)229-2763

Laboratory Job ID: 190-25693-1

Client Project/Site: WWTP - SEMIANNUAL PFAS

For:

City of Cadillac Utilities
1121 Plett Road
Cadillac, Michigan 49601

Attn: Cindy Tomaszewski

Sue Schafer

Authorized for release by:
4/27/2021 11:21:00 AM

Sue Schafer, Project Manager II
(810)229-2763
Sue.Schafer@Eurofinset.com

LINKS

Review your project
results through

TotalAccess

Have a Question?



Visit us at:

www.eurofinsus.com/Env

This report has been electronically signed and authorized by the signatory. Electronic signature is intended to be the legally binding equivalent of a traditionally handwritten signature.

Results relate only to the items tested and the sample(s) as received by the laboratory.

Table of Contents

Cover Page	1
Table of Contents	2
Sample Summary	3
Case Narrative	4
Client Sample Results	5
QC Sample Results	11
Isotope Dilution Summary	15
Definitions/Glossary	16
QC Association Summary	17
Lab Chronicle	18
Certification Summary	19
Chain of Custody	20



Sample Summary

Client: City of Cadillac Utilities
Project/Site: WWTP - SEMIANNUAL PFAS

Job ID: 190-25693-1

Lab Sample ID	Client Sample ID	Matrix	Collected	Received	Asset ID
190-25693-1	EFFLUENT	Water	04/13/21 13:55	04/15/21 08:00	
190-25693-2	EFFLUENT DUPLICATE	Water	04/13/21 14:00	04/15/21 08:00	
190-25693-3	FIELD BLANK	Water	04/13/21 13:50	04/15/21 08:00	

Case Narrative

Client: City of Cadillac Utilities
Project/Site: WWTP - SEMIANNUAL PFAS

Job ID: 190-25693-1

Job ID: 190-25693-1

Laboratory: Eurofins TestAmerica, Michigan

Narrative

Job Narrative 190-25693-1

Comments

No additional comments.

Receipt

The samples were received on 4/15/2021 8:00 AM. Unless otherwise noted below, the samples arrived in good condition, and where required, properly preserved and on ice. The temperature of the cooler at receipt was 2.0° C.

LCMS

Method 537 (modified): 13C2 PFTeDA Isotope Dilution Analyte (IDA) recovery associated with the following samples is below the method recommended limit: EFFLUENT (190-25693-1) and EFFLUENT DUPLICATE (190-25693-2). Generally, data quality is not considered affected if the IDA signal-to-noise ratio is greater than 10:1, which is achieved for all IDA in the samples.

No additional analytical or quality issues were noted, other than those described above or in the Definitions/Glossary page.

Client Sample Results

Client: City of Cadillac Utilities
Project/Site: WWTP - SEMI ANNUAL PFAS

Job ID: 190-25693-1

Client Sample ID: EFFLUENT

Lab Sample ID: 190-25693-1

Date Collected: 04/13/21 13:55

Matrix: Water

Date Received: 04/15/21 08:00

Method: 537 (modified) - Fluorinated Alkyl Substances

Analyte	Result	Qualifier	RL	Unit	D	Prepared	Analyzed	Dil Fac
11-Chloroeicosafluoro-3-oxaundecan e-1-sulfonic acid	<1.6		1.6	ng/L		04/23/21 12:56	04/23/21 20:11	1
9-Chlorohexadecafluoro-3-oxanonan e-1-sulfonic acid	<1.6		1.6	ng/L		04/23/21 12:56	04/23/21 20:11	1
DONA	<1.6		1.6	ng/L		04/23/21 12:56	04/23/21 20:11	1
4:2 FTS	<1.6		1.6	ng/L		04/23/21 12:56	04/23/21 20:11	1
6:2 FTS	<4.0		4.0	ng/L		04/23/21 12:56	04/23/21 20:11	1
8:2 FTS	<1.6		1.6	ng/L		04/23/21 12:56	04/23/21 20:11	1
HFPO-DA	<3.2		3.2	ng/L		04/23/21 12:56	04/23/21 20:11	1
N-ethylperfluorooctanesulfonamidoac etic acid (NEtFOSAA)	<4.0		4.0	ng/L		04/23/21 12:56	04/23/21 20:11	1
N-methylperfluorooctanesulfonamidoa cetic acid (NMeFOSAA)	<4.0		4.0	ng/L		04/23/21 12:56	04/23/21 20:11	1
Perfluorobutanesulfonic acid (PFBS)	19		1.6	ng/L		04/23/21 12:56	04/23/21 20:11	1
Perfluorobutanoic acid (PFBA)	<4.0		4.0	ng/L		04/23/21 12:56	04/23/21 20:11	1
Perfluorodecanesulfonic acid (PFDS)	<1.6		1.6	ng/L		04/23/21 12:56	04/23/21 20:11	1
Perfluorodecanoic acid (PFDA)	<1.6		1.6	ng/L		04/23/21 12:56	04/23/21 20:11	1
Perfluorododecanoic acid (PFDoA)	<1.6		1.6	ng/L		04/23/21 12:56	04/23/21 20:11	1
Perfluoroheptanesulfonic Acid (PFHpS)	<1.6		1.6	ng/L		04/23/21 12:56	04/23/21 20:11	1
Perfluoroheptanoic acid (PFHpA)	<1.6		1.6	ng/L		04/23/21 12:56	04/23/21 20:11	1
Perfluorohexanesulfonic acid (PFHxS)	<1.6		1.6	ng/L		04/23/21 12:56	04/23/21 20:11	1
Perfluorohexanoic acid (PFHxA)	11		1.6	ng/L		04/23/21 12:56	04/23/21 20:11	1
Perfluorononanesulfonic acid (PFNS)	<1.6		1.6	ng/L		04/23/21 12:56	04/23/21 20:11	1
Perfluorononanoic acid (PFNA)	<1.6		1.6	ng/L		04/23/21 12:56	04/23/21 20:11	1
Perfluorooctanesulfonamide (FOSA)	<1.6		1.6	ng/L		04/23/21 12:56	04/23/21 20:11	1
Perfluorooctane sulfonate (PFOS)	<1.6		1.6	ng/L		04/23/21 12:56	04/23/21 20:11	1
Perfluorooctanoic acid (PFOA)	1.9		1.6	ng/L		04/23/21 12:56	04/23/21 20:11	1
Perfluoropentanesulfonic acid (PFPeS)	<1.6		1.6	ng/L		04/23/21 12:56	04/23/21 20:11	1
Perfluoropentanoic acid (PFPeA)	2.6		1.6	ng/L		04/23/21 12:56	04/23/21 20:11	1
Perfluorotetradecanoic acid (PFTeA)	<1.6		1.6	ng/L		04/23/21 12:56	04/23/21 20:11	1
Perfluorotridecanoic acid (PFTriA)	<1.6		1.6	ng/L		04/23/21 12:56	04/23/21 20:11	1
Perfluoroundecanoic acid (PFUnA)	<1.6		1.6	ng/L		04/23/21 12:56	04/23/21 20:11	1

Isotope Dilution	%Recovery	Qualifier	Limits	Prepared	Analyzed	Dil Fac
13C8 FOSA	75		25 - 150	04/23/21 12:56	04/23/21 20:11	1
13C3 HFPO-DA	84		25 - 150	04/23/21 12:56	04/23/21 20:11	1
13C4 PFBA	89		25 - 150	04/23/21 12:56	04/23/21 20:11	1
13C3 PFBS	78		50 - 150	04/23/21 12:56	04/23/21 20:11	1
13C2 PFDA	100		50 - 150	04/23/21 12:56	04/23/21 20:11	1
13C2 PFDoA	91		50 - 150	04/23/21 12:56	04/23/21 20:11	1
13C4 PFHpA	95		50 - 150	04/23/21 12:56	04/23/21 20:11	1
13C2 PFHxA	86		50 - 150	04/23/21 12:56	04/23/21 20:11	1
13C5 PFNA	104		50 - 150	04/23/21 12:56	04/23/21 20:11	1
13C4 PFOA	97		50 - 150	04/23/21 12:56	04/23/21 20:11	1
13C4 PFOS	85		50 - 150	04/23/21 12:56	04/23/21 20:11	1
13C5 PFPeA	85		25 - 150	04/23/21 12:56	04/23/21 20:11	1
13C2 PFTeDA	48	*5-	50 - 150	04/23/21 12:56	04/23/21 20:11	1
13C2 PFUnA	100		50 - 150	04/23/21 12:56	04/23/21 20:11	1
d5-NEtFOSAA	98		50 - 150	04/23/21 12:56	04/23/21 20:11	1

Eurofins TestAmerica, Michigan

Client Sample Results

Client: City of Cadillac Utilities
Project/Site: WWTP - SEMIANNUAL PFAS

Job ID: 190-25693-1

Client Sample ID: EFFLUENT

Lab Sample ID: 190-25693-1

Date Collected: 04/13/21 13:55

Matrix: Water

Date Received: 04/15/21 08:00

Method: 537 (modified) - Fluorinated Alkyl Substances (Continued)

<i>Isotope Dilution</i>	<i>%Recovery</i>	<i>Qualifier</i>	<i>Limits</i>	<i>Prepared</i>	<i>Analyzed</i>	<i>Dil Fac</i>
d3-NMeFOSAA	89		50 - 150	04/23/21 12:56	04/23/21 20:11	1
M2-4:2 FTS	117		25 - 150	04/23/21 12:56	04/23/21 20:11	1
M2-6:2 FTS	137		25 - 150	04/23/21 12:56	04/23/21 20:11	1
M2-8:2 FTS	129		25 - 150	04/23/21 12:56	04/23/21 20:11	1
18O2 PFHxS	83		50 - 150	04/23/21 12:56	04/23/21 20:11	1

Client Sample Results

Client: City of Cadillac Utilities
Project/Site: WWTP - SEMI ANNUAL PFAS

Job ID: 190-25693-1

Client Sample ID: EFFLUENT DUPLICATE

Lab Sample ID: 190-25693-2

Date Collected: 04/13/21 14:00

Matrix: Water

Date Received: 04/15/21 08:00

Method: 537 (modified) - Fluorinated Alkyl Substances

Analyte	Result	Qualifier	RL	Unit	D	Prepared	Analyzed	Dil Fac
11-Chloroeicosafluoro-3-oxaundecan e-1-sulfonic acid	<1.6		1.6	ng/L		04/23/21 12:56	04/23/21 20:19	1
9-Chlorohexadecafluoro-3-oxanonan e-1-sulfonic acid	<1.6		1.6	ng/L		04/23/21 12:56	04/23/21 20:19	1
DONA	<1.6		1.6	ng/L		04/23/21 12:56	04/23/21 20:19	1
4:2 FTS	<1.6		1.6	ng/L		04/23/21 12:56	04/23/21 20:19	1
6:2 FTS	<4.0		4.0	ng/L		04/23/21 12:56	04/23/21 20:19	1
8:2 FTS	<1.6		1.6	ng/L		04/23/21 12:56	04/23/21 20:19	1
HFPO-DA	<3.2		3.2	ng/L		04/23/21 12:56	04/23/21 20:19	1
N-ethylperfluorooctanesulfonamidoac etic acid (NEtFOSAA)	<4.0		4.0	ng/L		04/23/21 12:56	04/23/21 20:19	1
N-methylperfluorooctanesulfonamidoa cetic acid (NMeFOSAA)	<4.0		4.0	ng/L		04/23/21 12:56	04/23/21 20:19	1
Perfluorobutanesulfonic acid (PFBS)	18		1.6	ng/L		04/23/21 12:56	04/23/21 20:19	1
Perfluorobutanoic acid (PFBA)	<4.0		4.0	ng/L		04/23/21 12:56	04/23/21 20:19	1
Perfluorodecanesulfonic acid (PFDS)	<1.6		1.6	ng/L		04/23/21 12:56	04/23/21 20:19	1
Perfluorodecanoic acid (PFDA)	<1.6		1.6	ng/L		04/23/21 12:56	04/23/21 20:19	1
Perfluorododecanoic acid (PFDoA)	<1.6		1.6	ng/L		04/23/21 12:56	04/23/21 20:19	1
Perfluoroheptanesulfonic Acid (PFHpS)	<1.6		1.6	ng/L		04/23/21 12:56	04/23/21 20:19	1
Perfluoroheptanoic acid (PFHpA)	<1.6		1.6	ng/L		04/23/21 12:56	04/23/21 20:19	1
Perfluorohexanesulfonic acid (PFHxS)	<1.6		1.6	ng/L		04/23/21 12:56	04/23/21 20:19	1
Perfluorohexanoic acid (PFHxA)	10		1.6	ng/L		04/23/21 12:56	04/23/21 20:19	1
Perfluorononanesulfonic acid (PFNS)	<1.6		1.6	ng/L		04/23/21 12:56	04/23/21 20:19	1
Perfluorononanoic acid (PFNA)	<1.6		1.6	ng/L		04/23/21 12:56	04/23/21 20:19	1
Perfluorooctanesulfonamide (FOSA)	<1.6		1.6	ng/L		04/23/21 12:56	04/23/21 20:19	1
Perfluorooctane sulfonate (PFOS)	<1.6		1.6	ng/L		04/23/21 12:56	04/23/21 20:19	1
Perfluorooctanoic acid (PFOA)	1.9		1.6	ng/L		04/23/21 12:56	04/23/21 20:19	1
Perfluoropentanesulfonic acid (PFPeS)	<1.6		1.6	ng/L		04/23/21 12:56	04/23/21 20:19	1
Perfluoropentanoic acid (PFPeA)	2.5		1.6	ng/L		04/23/21 12:56	04/23/21 20:19	1
Perfluorotetradecanoic acid (PFTeA)	<1.6		1.6	ng/L		04/23/21 12:56	04/23/21 20:19	1
Perfluorotridecanoic acid (PFTriA)	<1.6		1.6	ng/L		04/23/21 12:56	04/23/21 20:19	1
Perfluoroundecanoic acid (PFUnA)	<1.6		1.6	ng/L		04/23/21 12:56	04/23/21 20:19	1

Isotope Dilution	%Recovery	Qualifier	Limits	Prepared	Analyzed	Dil Fac
13C8 FOSA	74		25 - 150	04/23/21 12:56	04/23/21 20:19	1
13C3 HFPO-DA	81		25 - 150	04/23/21 12:56	04/23/21 20:19	1
13C4 PFBA	94		25 - 150	04/23/21 12:56	04/23/21 20:19	1
13C3 PFBS	83		50 - 150	04/23/21 12:56	04/23/21 20:19	1
13C2 PFDA	97		50 - 150	04/23/21 12:56	04/23/21 20:19	1
13C2 PFDoA	89		50 - 150	04/23/21 12:56	04/23/21 20:19	1
13C4 PFHpA	94		50 - 150	04/23/21 12:56	04/23/21 20:19	1
13C2 PFHxA	94		50 - 150	04/23/21 12:56	04/23/21 20:19	1
13C5 PFNA	106		50 - 150	04/23/21 12:56	04/23/21 20:19	1
13C4 PFOA	100		50 - 150	04/23/21 12:56	04/23/21 20:19	1
13C4 PFOS	82		50 - 150	04/23/21 12:56	04/23/21 20:19	1
13C5 PFPeA	90		25 - 150	04/23/21 12:56	04/23/21 20:19	1
13C2 PFTeA	47	*5-	50 - 150	04/23/21 12:56	04/23/21 20:19	1
13C2 PFUnA	109		50 - 150	04/23/21 12:56	04/23/21 20:19	1
d5-NEtFOSAA	92		50 - 150	04/23/21 12:56	04/23/21 20:19	1

Eurofins TestAmerica, Michigan

Client Sample Results

Client: City of Cadillac Utilities
Project/Site: WWTP - SEMIANNUAL PFAS

Job ID: 190-25693-1

Client Sample ID: EFFLUENT DUPLICATE

Lab Sample ID: 190-25693-2

Date Collected: 04/13/21 14:00

Matrix: Water

Date Received: 04/15/21 08:00

Method: 537 (modified) - Fluorinated Alkyl Substances (Continued)

<i>Isotope Dilution</i>	<i>%Recovery</i>	<i>Qualifier</i>	<i>Limits</i>	<i>Prepared</i>	<i>Analyzed</i>	<i>Dil Fac</i>
d3-NMeFOSAA	87		50 - 150	04/23/21 12:56	04/23/21 20:19	1
M2-4:2 FTS	121		25 - 150	04/23/21 12:56	04/23/21 20:19	1
M2-6:2 FTS	129		25 - 150	04/23/21 12:56	04/23/21 20:19	1
M2-8:2 FTS	142		25 - 150	04/23/21 12:56	04/23/21 20:19	1
18O2 PFHxS	84		50 - 150	04/23/21 12:56	04/23/21 20:19	1

Client Sample Results

Client: City of Cadillac Utilities
Project/Site: WWTP - SEMIANNUAL PFAS

Job ID: 190-25693-1

Client Sample ID: FIELD BLANK

Lab Sample ID: 190-25693-3

Date Collected: 04/13/21 13:50

Matrix: Water

Date Received: 04/15/21 08:00

Method: 537 (modified) - Fluorinated Alkyl Substances

Analyte	Result	Qualifier	RL	Unit	D	Prepared	Analyzed	Dil Fac
11-Chloroeicosafluoro-3-oxaundecan e-1-sulfonic acid	<1.8		1.8	ng/L		04/23/21 12:56	04/23/21 20:27	1
9-Chlorohexadecafluoro-3-oxanonan e-1-sulfonic acid	<1.8		1.8	ng/L		04/23/21 12:56	04/23/21 20:27	1
DONA	<1.8		1.8	ng/L		04/23/21 12:56	04/23/21 20:27	1
4:2 FTS	<1.8		1.8	ng/L		04/23/21 12:56	04/23/21 20:27	1
6:2 FTS	<4.4		4.4	ng/L		04/23/21 12:56	04/23/21 20:27	1
8:2 FTS	<1.8		1.8	ng/L		04/23/21 12:56	04/23/21 20:27	1
HFPO-DA	<3.5		3.5	ng/L		04/23/21 12:56	04/23/21 20:27	1
N-ethylperfluorooctanesulfonamidoac etic acid (NEtFOSAA)	<4.4		4.4	ng/L		04/23/21 12:56	04/23/21 20:27	1
N-methylperfluorooctanesulfonamidoa cetic acid (NMeFOSAA)	<4.4		4.4	ng/L		04/23/21 12:56	04/23/21 20:27	1
Perfluorobutanesulfonic acid (PFBS)	<1.8		1.8	ng/L		04/23/21 12:56	04/23/21 20:27	1
Perfluorobutanoic acid (PFBA)	<4.4		4.4	ng/L		04/23/21 12:56	04/23/21 20:27	1
Perfluorodecanesulfonic acid (PFDS)	<1.8		1.8	ng/L		04/23/21 12:56	04/23/21 20:27	1
Perfluorodecanoic acid (PFDA)	<1.8		1.8	ng/L		04/23/21 12:56	04/23/21 20:27	1
Perfluorododecanoic acid (PFDoA)	<1.8		1.8	ng/L		04/23/21 12:56	04/23/21 20:27	1
Perfluoroheptanesulfonic Acid (PFHpS)	<1.8		1.8	ng/L		04/23/21 12:56	04/23/21 20:27	1
Perfluoroheptanoic acid (PFHpA)	<1.8		1.8	ng/L		04/23/21 12:56	04/23/21 20:27	1
Perfluorohexanesulfonic acid (PFHxS)	<1.8		1.8	ng/L		04/23/21 12:56	04/23/21 20:27	1
Perfluorohexanoic acid (PFHxA)	<1.8		1.8	ng/L		04/23/21 12:56	04/23/21 20:27	1
Perfluorononanesulfonic acid (PFNS)	<1.8		1.8	ng/L		04/23/21 12:56	04/23/21 20:27	1
Perfluorononanoic acid (PFNA)	<1.8		1.8	ng/L		04/23/21 12:56	04/23/21 20:27	1
Perfluorooctanesulfonamide (FOSA)	<1.8		1.8	ng/L		04/23/21 12:56	04/23/21 20:27	1
Perfluorooctane sulfonate (PFOS)	<1.8		1.8	ng/L		04/23/21 12:56	04/23/21 20:27	1
Perfluorooctanoic acid (PFOA)	<1.8		1.8	ng/L		04/23/21 12:56	04/23/21 20:27	1
Perfluoropentanesulfonic acid (PFPeS)	<1.8		1.8	ng/L		04/23/21 12:56	04/23/21 20:27	1
Perfluoropentanoic acid (PFPeA)	<1.8		1.8	ng/L		04/23/21 12:56	04/23/21 20:27	1
Perfluorotetradecanoic acid (PFTeA)	<1.8		1.8	ng/L		04/23/21 12:56	04/23/21 20:27	1
Perfluorotridecanoic acid (PFTriA)	<1.8		1.8	ng/L		04/23/21 12:56	04/23/21 20:27	1
Perfluoroundecanoic acid (PFUnA)	<1.8		1.8	ng/L		04/23/21 12:56	04/23/21 20:27	1

Isotope Dilution	%Recovery	Qualifier	Limits	Prepared	Analyzed	Dil Fac
13C8 FOSA	71		25 - 150	04/23/21 12:56	04/23/21 20:27	1
13C3 HFPO-DA	98		25 - 150	04/23/21 12:56	04/23/21 20:27	1
13C4 PFBA	91		25 - 150	04/23/21 12:56	04/23/21 20:27	1
13C3 PFBS	83		50 - 150	04/23/21 12:56	04/23/21 20:27	1
13C2 PFDA	99		50 - 150	04/23/21 12:56	04/23/21 20:27	1
13C2 PFDoA	88		50 - 150	04/23/21 12:56	04/23/21 20:27	1
13C4 PFHpA	100		50 - 150	04/23/21 12:56	04/23/21 20:27	1
13C2 PFHxA	96		50 - 150	04/23/21 12:56	04/23/21 20:27	1
13C5 PFNA	99		50 - 150	04/23/21 12:56	04/23/21 20:27	1
13C4 PFOA	99		50 - 150	04/23/21 12:56	04/23/21 20:27	1
13C4 PFOS	87		50 - 150	04/23/21 12:56	04/23/21 20:27	1
13C5 PFPeA	90		25 - 150	04/23/21 12:56	04/23/21 20:27	1
13C2 PFTeDA	86		50 - 150	04/23/21 12:56	04/23/21 20:27	1
13C2 PFUnA	88		50 - 150	04/23/21 12:56	04/23/21 20:27	1
d5-NEtFOSAA	89		50 - 150	04/23/21 12:56	04/23/21 20:27	1
d3-NMeFOSAA	89		50 - 150	04/23/21 12:56	04/23/21 20:27	1

Eurofins TestAmerica, Michigan

Client Sample Results

Client: City of Cadillac Utilities
Project/Site: WWTP - SEMIANNUAL PFAS

Job ID: 190-25693-1

Client Sample ID: FIELD BLANK

Lab Sample ID: 190-25693-3

Date Collected: 04/13/21 13:50

Matrix: Water

Date Received: 04/15/21 08:00

Method: 537 (modified) - Fluorinated Alkyl Substances (Continued)

<i>Isotope Dilution</i>	<i>%Recovery</i>	<i>Qualifier</i>	<i>Limits</i>	<i>Prepared</i>	<i>Analyzed</i>	<i>Dil Fac</i>
M2-4:2 FTS	107		25 - 150	04/23/21 12:56	04/23/21 20:27	1
M2-6:2 FTS	110		25 - 150	04/23/21 12:56	04/23/21 20:27	1
M2-8:2 FTS	119		25 - 150	04/23/21 12:56	04/23/21 20:27	1
18O2 PFHxS	85		50 - 150	04/23/21 12:56	04/23/21 20:27	1

QC Sample Results

Client: City of Cadillac Utilities
Project/Site: WWTP - SEMIANNUAL PFAS

Job ID: 190-25693-1

Method: 537 (modified) - Fluorinated Alkyl Substances

Lab Sample ID: MB 200-166136/1-A

Matrix: Water

Analysis Batch: 166148

Client Sample ID: Method Blank

Prep Type: Total/NA

Prep Batch: 166136

Analyte	MB Result	MB Qualifier	RL	Unit	D	Prepared	Analyzed	Dil Fac
11-Chloroeicosafluoro-3-oxaundecan e-1-sulfonic acid	<2.0		2.0	ng/L		04/23/21 12:56	04/23/21 17:41	1
9-Chlorohexadecafluoro-3-oxanonan e-1-sulfonic acid	<2.0		2.0	ng/L		04/23/21 12:56	04/23/21 17:41	1
DONA	<2.0		2.0	ng/L		04/23/21 12:56	04/23/21 17:41	1
4:2 FTS	<2.0		2.0	ng/L		04/23/21 12:56	04/23/21 17:41	1
6:2 FTS	<5.0		5.0	ng/L		04/23/21 12:56	04/23/21 17:41	1
8:2 FTS	<2.0		2.0	ng/L		04/23/21 12:56	04/23/21 17:41	1
HFPO-DA	<4.0		4.0	ng/L		04/23/21 12:56	04/23/21 17:41	1
N-ethylperfluorooctanesulfonamidoac etic acid (NEtFOSAA)	<5.0		5.0	ng/L		04/23/21 12:56	04/23/21 17:41	1
N-methylperfluorooctanesulfonamidoa cetic acid (NMeFOSAA)	<5.0		5.0	ng/L		04/23/21 12:56	04/23/21 17:41	1
Perfluorobutanesulfonic acid (PFBS)	<2.0		2.0	ng/L		04/23/21 12:56	04/23/21 17:41	1
Perfluorobutanoic acid (PFBA)	<5.0		5.0	ng/L		04/23/21 12:56	04/23/21 17:41	1
Perfluorodecanesulfonic acid (PFDS)	<2.0		2.0	ng/L		04/23/21 12:56	04/23/21 17:41	1
Perfluorodecanoic acid (PFDA)	<2.0		2.0	ng/L		04/23/21 12:56	04/23/21 17:41	1
Perfluorododecanoic acid (PFDoA)	<2.0		2.0	ng/L		04/23/21 12:56	04/23/21 17:41	1
Perfluoroheptanesulfonic Acid (PFHpS)	<2.0		2.0	ng/L		04/23/21 12:56	04/23/21 17:41	1
Perfluoroheptanoic acid (PFHpA)	<2.0		2.0	ng/L		04/23/21 12:56	04/23/21 17:41	1
Perfluorohexanesulfonic acid (PFHxS)	<2.0		2.0	ng/L		04/23/21 12:56	04/23/21 17:41	1
Perfluorohexanoic acid (PFHxA)	<2.0		2.0	ng/L		04/23/21 12:56	04/23/21 17:41	1
Perfluorononanesulfonic acid (PFNS)	<2.0		2.0	ng/L		04/23/21 12:56	04/23/21 17:41	1
Perfluorononanoic acid (PFNA)	<2.0		2.0	ng/L		04/23/21 12:56	04/23/21 17:41	1
Perfluorooctanesulfonamide (FOSA)	<2.0		2.0	ng/L		04/23/21 12:56	04/23/21 17:41	1
Perfluorooctane sulfonate (PFOS)	<2.0		2.0	ng/L		04/23/21 12:56	04/23/21 17:41	1
Perfluorooctanoic acid (PFOA)	<2.0		2.0	ng/L		04/23/21 12:56	04/23/21 17:41	1
Perfluoropentanesulfonic acid (PFPeS)	<2.0		2.0	ng/L		04/23/21 12:56	04/23/21 17:41	1
Perfluoropentanoic acid (PFPeA)	<2.0		2.0	ng/L		04/23/21 12:56	04/23/21 17:41	1
Perfluorotetradecanoic acid (PFTeA)	<2.0		2.0	ng/L		04/23/21 12:56	04/23/21 17:41	1
Perfluorotridecanoic acid (PFTriA)	<2.0		2.0	ng/L		04/23/21 12:56	04/23/21 17:41	1
Perfluoroundecanoic acid (PFUnA)	<2.0		2.0	ng/L		04/23/21 12:56	04/23/21 17:41	1

Isotope Dilution	MB %Recovery	MB Qualifier	Limits	Prepared	Analyzed	Dil Fac
13C8 FOSA	74		25 - 150	04/23/21 12:56	04/23/21 17:41	1
13C3 HFPO-DA	82		25 - 150	04/23/21 12:56	04/23/21 17:41	1
13C4 PFBA	100		25 - 150	04/23/21 12:56	04/23/21 17:41	1
13C3 PFBS	94		50 - 150	04/23/21 12:56	04/23/21 17:41	1
13C2 PFDA	95		50 - 150	04/23/21 12:56	04/23/21 17:41	1
13C2 PFDoA	82		50 - 150	04/23/21 12:56	04/23/21 17:41	1
13C4 PFHpA	96		50 - 150	04/23/21 12:56	04/23/21 17:41	1
13C2 PFHxA	100		50 - 150	04/23/21 12:56	04/23/21 17:41	1
13C5 PFNA	98		50 - 150	04/23/21 12:56	04/23/21 17:41	1
13C4 PFOA	95		50 - 150	04/23/21 12:56	04/23/21 17:41	1
13C4 PFOS	92		50 - 150	04/23/21 12:56	04/23/21 17:41	1
13C5 PFPeA	98		25 - 150	04/23/21 12:56	04/23/21 17:41	1
13C2 PFTeDA	82		50 - 150	04/23/21 12:56	04/23/21 17:41	1
13C2 PFUnA	93		50 - 150	04/23/21 12:56	04/23/21 17:41	1

Eurofins TestAmerica, Michigan

QC Sample Results

Client: City of Cadillac Utilities
Project/Site: WWTP - SEMIANNUAL PFAS

Job ID: 190-25693-1

Method: 537 (modified) - Fluorinated Alkyl Substances (Continued)

Lab Sample ID: MB 200-166136/1-A

Matrix: Water

Analysis Batch: 166148

Client Sample ID: Method Blank

Prep Type: Total/NA

Prep Batch: 166136

Isotope Dilution	MB %Recovery	MB Qualifier	Limits	Prepared	Analyzed	Dil Fac
d5-NEtFOSAA	96		50 - 150	04/23/21 12:56	04/23/21 17:41	1
d3-NMeFOSAA	90		50 - 150	04/23/21 12:56	04/23/21 17:41	1
M2-4:2 FTS	102		25 - 150	04/23/21 12:56	04/23/21 17:41	1
M2-6:2 FTS	98		25 - 150	04/23/21 12:56	04/23/21 17:41	1
M2-8:2 FTS	101		25 - 150	04/23/21 12:56	04/23/21 17:41	1
18O2 PFHxS	91		50 - 150	04/23/21 12:56	04/23/21 17:41	1

Lab Sample ID: LCS 200-166136/2-A

Matrix: Water

Analysis Batch: 166148

Client Sample ID: Lab Control Sample

Prep Type: Total/NA

Prep Batch: 166136

Analyte	Spike Added	LCS Result	LCS Qualifier	Unit	D	%Rec	%Rec. Limits
11-Chloroeicosafluoro-3-oxaundecane-1-sulfonic acid	37.7	31.1		ng/L		83	50 - 150
9-Chlorohexadecafluoro-3-oxanonane-1-sulfonic acid	37.3	36.8		ng/L		99	50 - 150
DONA	37.7	40.1		ng/L		107	50 - 150
4:2 FTS	37.4	39.4		ng/L		105	50 - 150
6:2 FTS	37.9	41.4		ng/L		109	50 - 150
8:2 FTS	38.3	40.4		ng/L		105	50 - 150
HFPO-DA	40.0	39.8		ng/L		100	50 - 150
N-ethylperfluorooctanesulfonamidoacetic acid (NEtFOSAA)	40.0	38.3		ng/L		96	70 - 130
N-methylperfluorooctanesulfonamidoacetic acid (NMeFOSAA)	40.0	36.2		ng/L		91	70 - 130
Perfluorobutanesulfonic acid (PFBS)	35.4	35.0		ng/L		99	70 - 130
Perfluorobutanoic acid (PFBA)	40.0	39.5		ng/L		99	50 - 150
Perfluorodecanesulfonic acid (PFDS)	38.6	32.7		ng/L		85	50 - 150
Perfluorodecanoic acid (PFDA)	40.0	41.0		ng/L		102	70 - 130
Perfluorododecanoic acid (PFDoA)	40.0	40.3		ng/L		101	70 - 130
Perfluoroheptanesulfonic Acid (PFHpS)	38.1	36.9		ng/L		97	50 - 150
Perfluoroheptanoic acid (PFHpA)	40.0	41.3		ng/L		103	70 - 130
Perfluorohexanesulfonic acid (PFHxS)	36.4	35.6		ng/L		98	70 - 130
Perfluorohexanoic acid (PFHxA)	40.0	39.6		ng/L		99	70 - 130
Perfluorononanesulfonic acid (PFNS)	38.4	36.8		ng/L		96	50 - 150
Perfluorononanoic acid (PFNA)	40.0	35.4		ng/L		88	70 - 130
Perfluorooctanesulfonamide (FOSA)	40.0	40.5		ng/L		101	50 - 150
Perfluorooctane sulfonate (PFOS)	37.1	36.8		ng/L		99	70 - 130
Perfluorooctanoic acid (PFOA)	40.0	39.8		ng/L		100	70 - 130
Perfluoropentanesulfonic acid (PFPeS)	37.5	38.1		ng/L		102	50 - 150
Perfluoropentanoic acid (PFPeA)	40.0	40.2		ng/L		100	50 - 150
Perfluorotetradecanoic acid (PFTeA)	40.0	38.4		ng/L		96	70 - 130

Eurofins TestAmerica, Michigan

QC Sample Results

Client: City of Cadillac Utilities
Project/Site: WWTP - SEMI-ANNUAL PFAS

Job ID: 190-25693-1

Method: 537 (modified) - Fluorinated Alkyl Substances (Continued)

Lab Sample ID: LCS 200-166136/2-A

Matrix: Water

Analysis Batch: 166148

Client Sample ID: Lab Control Sample

Prep Type: Total/NA

Prep Batch: 166136

Analyte			Spike	LCS	LCS	Unit	D	%Rec.		
			Added	Result	Qualifier			Limits		
Perfluorotridecanoic acid (PFTriA)			40.0	37.0		ng/L		93	70 - 130	
Perfluoroundecanoic acid (PFUnA)			40.0	38.9		ng/L		97	70 - 130	
		LCS	LCS							
Isotope Dilution	%Recovery	Qualifier	Limits							
13C8 FOSA	78		25 - 150							
13C3 HFPO-DA	97		25 - 150							
13C4 PFBA	104		25 - 150							
13C3 PFBS	95		50 - 150							
13C2 PFDA	94		50 - 150							
13C2 PFDoA	90		50 - 150							
13C4 PFHpA	103		50 - 150							
13C2 PFHxA	102		50 - 150							
13C5 PFNA	107		50 - 150							
13C4 PFOA	98		50 - 150							
13C4 PFOS	94		50 - 150							
13C5 PFPeA	103		25 - 150							
13C2 PFTeDA	88		50 - 150							
13C2 PFUnA	98		50 - 150							
d5-NEtFOSAA	101		50 - 150							
d3-NMeFOSAA	100		50 - 150							
M2-4:2 FTS	104		25 - 150							
M2-6:2 FTS	97		25 - 150							
M2-8:2 FTS	102		25 - 150							
18O2 PFHxS	93		50 - 150							

Lab Sample ID: LCSD 200-166136/3-A

Matrix: Water

Analysis Batch: 166148

Client Sample ID: Lab Control Sample Dup

Prep Type: Total/NA

Prep Batch: 166136

Analyte	Spike Added	LCSD Result	LCSD Qualifier	Unit	D	%Rec	%Rec. Limits	RPD	RPD Limit
11-Chloroeicosafluoro-3-oxaundecane-1-sulfonic acid	37.7	33.6		ng/L		89	50 - 150	8	30
9-Chlorohexadecafluoro-3-oxanonane-1-sulfonic acid	37.3	39.2		ng/L		105	50 - 150	6	30
DONA	37.7	39.4		ng/L		105	50 - 150	2	30
4:2 FTS	37.4	37.2		ng/L		100	50 - 150	6	30
6:2 FTS	37.9	42.0		ng/L		111	50 - 150	2	30
8:2 FTS	38.3	41.1		ng/L		107	50 - 150	2	30
HFPO-DA	40.0	35.5		ng/L		89	50 - 150	12	30
N-ethylperfluorooctanesulfonamidoacetic acid (NEtFOSAA)	40.0	40.7		ng/L		102	70 - 130	6	20
N-methylperfluorooctanesulfonamidoacetic acid (NMeFOSAA)	40.0	36.6		ng/L		92	70 - 130	1	20
Perfluorobutanesulfonic acid (PFBS)	35.4	36.0		ng/L		102	70 - 130	3	20
Perfluorobutanoic acid (PFBA)	40.0	40.6		ng/L		102	50 - 150	3	30
Perfluorodecanesulfonic acid (PFDS)	38.6	35.9		ng/L		93	50 - 150	9	30
Perfluorodecanoic acid (PFDA)	40.0	39.5		ng/L		99	70 - 130	4	20

Eurofins TestAmerica, Michigan

QC Sample Results

Client: City of Cadillac Utilities
Project/Site: WWTP - SEMI-ANNUAL PFAS

Job ID: 190-25693-1

Method: 537 (modified) - Fluorinated Alkyl Substances (Continued)

Lab Sample ID: LCSD 200-166136/3-A

Matrix: Water

Analysis Batch: 166148

Client Sample ID: Lab Control Sample Dup

Prep Type: Total/NA

Prep Batch: 166136

Analyte	Spike Added	LCSD Result	LCSD Qualifier	Unit	D	%Rec	%Rec. Limits	RPD	RPD Limit
Perfluorododecanoic acid (PFDoA)	40.0	41.8		ng/L		104	70 - 130	4	20
Perfluoroheptanesulfonic Acid (PFHpS)	38.1	36.9		ng/L		97	50 - 150	0	30
Perfluoroheptanoic acid (PFHpA)	40.0	41.9		ng/L		105	70 - 130	1	20
Perfluorohexanesulfonic acid (PFHxS)	36.4	36.4		ng/L		100	70 - 130	2	20
Perfluorohexanoic acid (PFHxA)	40.0	38.2		ng/L		95	70 - 130	4	20
Perfluorononanesulfonic acid (PFNS)	38.4	38.9		ng/L		101	50 - 150	5	30
Perfluorononanoic acid (PFNA)	40.0	35.9		ng/L		90	70 - 130	1	20
Perfluorooctanesulfonamide (FOSA)	40.0	42.2		ng/L		106	50 - 150	4	30
Perfluorooctane sulfonate (PFOS)	37.1	38.0		ng/L		102	70 - 130	3	20
Perfluorooctanoic acid (PFOA)	40.0	40.4		ng/L		101	70 - 130	1	20
Perfluoropentanesulfonic acid (PFPeS)	37.5	38.9		ng/L		104	50 - 150	2	30
Perfluoropentanoic acid (PFPeA)	40.0	40.8		ng/L		102	50 - 150	2	30
Perfluorotetradecanoic acid (PFTeA)	40.0	42.1		ng/L		105	70 - 130	9	20
Perfluorotridecanoic acid (PFTriA)	40.0	39.0		ng/L		97	70 - 130	5	20
Perfluoroundecanoic acid (PFUnA)	40.0	38.1		ng/L		95	70 - 130	2	20

Isotope Dilution	LCSD %Recovery	LCSD Qualifier	Limits
13C8 FOSA	75		25 - 150
13C3 HFPO-DA	98		25 - 150
13C4 PFBA	95		25 - 150
13C3 PFBS	88		50 - 150
13C2 PFDA	98		50 - 150
13C2 PFDoA	84		50 - 150
13C4 PFHpA	97		50 - 150
13C2 PFHxA	98		50 - 150
13C5 PFNA	101		50 - 150
13C4 PFOA	97		50 - 150
13C4 PFOS	91		50 - 150
13C5 PFPeA	95		25 - 150
13C2 PFTeDA	86		50 - 150
13C2 PFUnA	96		50 - 150
d5-NEtFOSAA	87		50 - 150
d3-NMeFOSAA	91		50 - 150
M2-4:2 FTS	94		25 - 150
M2-6:2 FTS	93		25 - 150
M2-8:2 FTS	103		25 - 150
18O2 PFHxS	89		50 - 150

Isotope Dilution Summary

Client: City of Cadillac Utilities
Project/Site: WWTP - SEMIANNUAL PFAS

Job ID: 190-25693-1

Method: 537 (modified) - Fluorinated Alkyl Substances

Matrix: Water

Prep Type: Total/NA

Percent Isotope Dilution Recovery (Acceptance Limits)

Lab Sample ID	Client Sample ID	PFOSA (25-150)	HFPODA (25-150)	PFBA (25-150)	C3PFBS (50-150)	PFDA (50-150)	PFDoA (50-150)	C4PFHA (50-150)	PFHxA (50-150)
190-25693-1	EFFLUENT	75	84	89	78	100	91	95	86
190-25693-2	EFFLUENT DUPLICATE	74	81	94	83	97	89	94	94
190-25693-3	FIELD BLANK	71	98	91	83	99	88	100	96
LCS 200-166136/2-A	Lab Control Sample	78	97	104	95	94	90	103	102
LCSD 200-166136/3-A	Lab Control Sample Dup	75	98	95	88	98	84	97	98
MB 200-166136/1-A	Method Blank	74	82	100	94	95	82	96	100

Percent Isotope Dilution Recovery (Acceptance Limits)

Lab Sample ID	Client Sample ID	PFNA (50-150)	PFOA (50-150)	PFOS (50-150)	PFPeA (25-150)	PFTDA (50-150)	PFUnA (50-150)	d5NEFOS (50-150)	d3NMFOS (50-150)
190-25693-1	EFFLUENT	104	97	85	85	48 *5-	100	98	89
190-25693-2	EFFLUENT DUPLICATE	106	100	82	90	47 *5-	109	92	87
190-25693-3	FIELD BLANK	99	99	87	90	86	88	89	89
LCS 200-166136/2-A	Lab Control Sample	107	98	94	103	88	98	101	100
LCSD 200-166136/3-A	Lab Control Sample Dup	101	97	91	95	86	96	87	91
MB 200-166136/1-A	Method Blank	98	95	92	98	82	93	96	90

Percent Isotope Dilution Recovery (Acceptance Limits)

Lab Sample ID	Client Sample ID	M242FTS (25-150)	M262FTS (25-150)	M282FTS (25-150)	PFHxS (50-150)
190-25693-1	EFFLUENT	117	137	129	83
190-25693-2	EFFLUENT DUPLICATE	121	129	142	84
190-25693-3	FIELD BLANK	107	110	119	85
LCS 200-166136/2-A	Lab Control Sample	104	97	102	93
LCSD 200-166136/3-A	Lab Control Sample Dup	94	93	103	89
MB 200-166136/1-A	Method Blank	102	98	101	91

Surrogate Legend

PFOSA = 13C8 FOSA
HFPODA = 13C3 HFPO-DA
PFBA = 13C4 PFBA
C3PFBS = 13C3 PFBS
PFDA = 13C2 PFDA
PFDoA = 13C2 PFDoA
C4PFHA = 13C4 PFHpA
PFHxA = 13C2 PFHxA
PFNA = 13C5 PFNA
PFOA = 13C4 PFOA
PFOS = 13C4 PFOS
PFPeA = 13C5 PFPeA
PFTDA = 13C2 PFTeDA
PFUnA = 13C2 PFUnA
d5NEFOS = d5-NEtFOSAA
d3NMFOS = d3-NMeFOSAA
M242FTS = M2-4:2 FTS
M262FTS = M2-6:2 FTS
M282FTS = M2-8:2 FTS
PFHxS = 18O2 PFHxS

Definitions/Glossary

Client: City of Cadillac Utilities
Project/Site: WWTP - SEMIANNUAL PFAS

Job ID: 190-25693-1

Qualifiers

LCMS

Qualifier	Qualifier Description
*5-	Isotope dilution analyte is outside acceptance limits, low biased.

Glossary

Abbreviation	These commonly used abbreviations may or may not be present in this report.
α	Listed under the "D" column to designate that the result is reported on a dry weight basis
%R	Percent Recovery
CFL	Contains Free Liquid
CFU	Colony Forming Unit
CNF	Contains No Free Liquid
DER	Duplicate Error Ratio (normalized absolute difference)
Dil Fac	Dilution Factor
DL	Detection Limit (DoD/DOE)
DL, RA, RE, IN	Indicates a Dilution, Re-analysis, Re-extraction, or additional Initial metals/anion analysis of the sample
DLC	Decision Level Concentration (Radiochemistry)
EDL	Estimated Detection Limit (Dioxin)
LOD	Limit of Detection (DoD/DOE)
LOQ	Limit of Quantitation (DoD/DOE)
MCL	EPA recommended "Maximum Contaminant Level"
MDA	Minimum Detectable Activity (Radiochemistry)
MDC	Minimum Detectable Concentration (Radiochemistry)
MDL	Method Detection Limit
ML	Minimum Level (Dioxin)
MPN	Most Probable Number
MQL	Method Quantitation Limit
NC	Not Calculated
ND	Not Detected at the reporting limit (or MDL or EDL if shown)
NEG	Negative / Absent
POS	Positive / Present
PQL	Practical Quantitation Limit
PRES	Presumptive
QC	Quality Control
RER	Relative Error Ratio (Radiochemistry)
RL	Reporting Limit or Requested Limit (Radiochemistry)
RPD	Relative Percent Difference, a measure of the relative difference between two points
TEF	Toxicity Equivalent Factor (Dioxin)
TEQ	Toxicity Equivalent Quotient (Dioxin)
TNTC	Too Numerous To Count

QC Association Summary

Client: City of Cadillac Utilities
Project/Site: WWTP - SEMIANNUAL PFAS

Job ID: 190-25693-1

LCMS

Prep Batch: 166136

Lab Sample ID	Client Sample ID	Prep Type	Matrix	Method	Prep Batch
190-25693-1	EFFLUENT	Total/NA	Water	3535	
190-25693-2	EFFLUENT DUPLICATE	Total/NA	Water	3535	
190-25693-3	FIELD BLANK	Total/NA	Water	3535	
MB 200-166136/1-A	Method Blank	Total/NA	Water	3535	
LCS 200-166136/2-A	Lab Control Sample	Total/NA	Water	3535	
LCSD 200-166136/3-A	Lab Control Sample Dup	Total/NA	Water	3535	

Analysis Batch: 166148

Lab Sample ID	Client Sample ID	Prep Type	Matrix	Method	Prep Batch
190-25693-1	EFFLUENT	Total/NA	Water	537 (modified)	166136
190-25693-2	EFFLUENT DUPLICATE	Total/NA	Water	537 (modified)	166136
190-25693-3	FIELD BLANK	Total/NA	Water	537 (modified)	166136
MB 200-166136/1-A	Method Blank	Total/NA	Water	537 (modified)	166136
LCS 200-166136/2-A	Lab Control Sample	Total/NA	Water	537 (modified)	166136
LCSD 200-166136/3-A	Lab Control Sample Dup	Total/NA	Water	537 (modified)	166136

Lab Chronicle

Client: City of Cadillac Utilities
Project/Site: WWTP - SEMIANNUAL PFAS

Job ID: 190-25693-1

Client Sample ID: EFFLUENT

Date Collected: 04/13/21 13:55

Date Received: 04/15/21 08:00

Lab Sample ID: 190-25693-1

Matrix: Water

Prep Type	Batch Type	Batch Method	Run	Dilution Factor	Batch Number	Prepared or Analyzed	Analyst	Lab
Total/NA	Prep	3535			166136	04/23/21 12:56	KFW	TAL BUR
Total/NA	Analysis	537 (modified)		1	166148	04/23/21 20:11	BWC	TAL BUR

Client Sample ID: EFFLUENT DUPLICATE

Date Collected: 04/13/21 14:00

Date Received: 04/15/21 08:00

Lab Sample ID: 190-25693-2

Matrix: Water

Prep Type	Batch Type	Batch Method	Run	Dilution Factor	Batch Number	Prepared or Analyzed	Analyst	Lab
Total/NA	Prep	3535			166136	04/23/21 12:56	KFW	TAL BUR
Total/NA	Analysis	537 (modified)		1	166148	04/23/21 20:19	BWC	TAL BUR

Client Sample ID: FIELD BLANK

Date Collected: 04/13/21 13:50

Date Received: 04/15/21 08:00

Lab Sample ID: 190-25693-3

Matrix: Water

Prep Type	Batch Type	Batch Method	Run	Dilution Factor	Batch Number	Prepared or Analyzed	Analyst	Lab
Total/NA	Prep	3535			166136	04/23/21 12:56	KFW	TAL BUR
Total/NA	Analysis	537 (modified)		1	166148	04/23/21 20:27	BWC	TAL BUR

Laboratory References:

TAL BUR = Eurofins TestAmerica, Burlington, 530 Community Drive, Suite 11, South Burlington, VT 05403, TEL (802)660-1990

Analyst References:

Lab: TAL BUR

Batch Type: Prep

KFW = Kurtis Wisniewski

Batch Type: Analysis

BWC = Bradley Chirgwin

Accreditation/Certification Summary

Client: City of Cadillac Utilities
Project/Site: WWTP - SEMIANNUAL PFAS

Job ID: 190-25693-1

Laboratory: Eurofins TestAmerica, Burlington

All accreditations/certifications held by this laboratory are listed. Not all accreditations/certifications are applicable to this report.

Authority	Program	Identification Number	Expiration Date
ANAB	Dept. of Defense ELAP	L2336	02-25-23
Connecticut	State	PH-0751	09-30-21
DE Haz. Subst. Cleanup Act (HSCA)	State	N/A	05-16-21
Florida	NELAP	E87467	06-30-21
Minnesota	NELAP	050-999-436	12-31-21
New Hampshire	NELAP	2006	12-18-21
New Jersey	NELAP	VT972	06-30-21
New York	NELAP	10391	04-01-22
Pennsylvania	NELAP	68-00489	04-30-22
Rhode Island	State	LAO00298	12-30-21
US Fish & Wildlife	US Federal Programs	058448	07-31-21
USDA	US Federal Programs	P330-17-00272	10-30-23
Vermont	State	VT4000	02-10-22
Virginia	NELAP	460209	12-14-21
Wisconsin	State	399133350	08-31-21

Eurofins TestAmerica Canton Sample Receipt Form/Narrative				Login # : <u>190-25693</u>	
Canton Facility					
Client <u>City of Cadillac</u>		Site Name _____		Cooler unpacked by: <u>Adam Jensen</u>	
Cooler Received on <u>4-15-21</u>		Opened on <u>4-15-21</u>			
FedEx: 1 st Grd Exp <u>UPS FAS Chopper</u>		Client Drop Off <u>TestAmerica Courier</u>		Other _____	
Receipt After-hours: Drop-off Date/Time			Storage Location		
TestAmerica Cooler # <u>74</u>		Foam Box _____		Client Cooler _____	
Packing material used: <u>Bubble Wrap</u>		Foam _____		Plastic Bag _____	
COOLANT: <u>Wet Ice</u>		Blue Ice _____		Dry Ice _____	
		Water _____		None _____	
1. Cooler temperature upon receipt				☐ See Multiple Cooler Form	
IR GUN# IR-11 (CF +0.1 °C)		Observed Cooler Temp. <u>19</u> °C		Corrected Cooler Temp. <u>20</u> °C	
IR GUN #IR-12 (CF +0.2 °C)		Observed Cooler Temp. _____ °C		Corrected Cooler Temp. _____ °C	
2. Were tamper/custody seals on the outside of the cooler(s)? If Yes Quantity <u>1</u>				☒ Yes ☐ No	
-Were the seals on the outside of the cooler(s) signed & dated?				☒ Yes ☐ No ☐ NA	
-Were tamper/custody seals on the bottle(s) or bottle kits (LLHg/MeHg)?				☒ Yes ☐ No ☐ NA	
-Were tamper/custody seals intact and uncompromised?				☒ Yes ☐ No ☐ NA	
3. Shippers' packing slip attached to the cooler(s)?				☒ Yes ☐ No	
4. Did custody papers accompany the sample(s)?				☒ Yes ☐ No	
5. Were the custody papers relinquished & signed in the appropriate place?				☒ Yes ☐ No	
6. Was/were the person(s) who collected the samples clearly identified on the COC?				☒ Yes ☐ No	
7. Did all bottles arrive in good condition (Unbroken)?				☒ Yes ☐ No	
8. Could all bottle labels (ID/Date/Time) be reconciled with the COC?				☒ Yes ☐ No	
9. For each sample, does the COC specify preservatives (Y/N), # of containers (Y/N), and sample type of grab/comp (Y/N)?					
10. Were correct bottle(s) used for the test(s) indicated?				☒ Yes ☐ No	
11. Sufficient quantity received to perform indicated analyses?				☒ Yes ☐ No	
12. Are these work share samples and all listed on the COC?				☒ Yes ☐ No	
If yes, Questions 13-17 have been checked at the originating laboratory.					
13. Were all preserved sample(s) at the correct pH upon receipt?				☐ Yes ☒ No ☐ NA pH Strip Lot# <u>HC022887</u>	
14. Were VOAs on the COC?				☐ Yes ☒ No ☐ NA	
15. Were air bubbles >6 mm in any VOA vials? Larger than this.				☐ Yes ☒ No ☐ NA	
16. Was a VOA trip blank present in the cooler(s)? Trip Blank Lot # _____				☐ Yes ☒ No	
17. Was a LL Hg or Me Hg trip blank present? _____				☐ Yes ☒ No	
Contacted PM _____ Date _____ by _____ via Verbal Voice Mail Other _____					
Concerning _____					

18. CHAIN OF CUSTODY & SAMPLE DISCREPANCIES		☐ additional next page	Samples processed by: _____
19. SAMPLE CONDITION			
Sample(s) _____ were received after the recommended holding time had expired.			
Sample(s) _____ were received in a broken container.			
Sample(s) _____ were received with bubble >6 mm in diameter. (Notify PM)			
20. SAMPLE PRESERVATION			
Sample(s) _____ were further preserved in the laboratory.			
Time preserved: _____ Preservative(s) added/Lot number(s): _____			
VOA Sample Preservation - Date/Time VOAs Frozen: _____			



Client Information (Sub Contract Lab)		Lab PM: Schafer, Sue	Carrier Tracking No(s): 240-134967.1
Client Contact: Shipping/Receiving		E-Mail: Sue.Schafer@Eurofinset.com	State of Origin: Michigan
Company: TestAmerica Laboratories, Inc.		Accreditations Required (See note): 190-25693-1	
Address: 530 Community Drive, Suite 11, City: South Burlington State, Zip: VT, 05403 Phone: 802-660-1990(Tel) 802-660-1919(Fax) Email: Project Name: WWTP Site:		Due Date Requested: 5/5/2021 TAT Requested (days): PO #: WO #: Project #: 19001884 SSOW#:	
Sample Identification - Client ID (Lab ID)		Analysis Requested	
EFFLUENT (190-25693-1)		Preservation Codes: A - HCL B - NaOH C - Zn Acetate D - Nitric Acid E - NaHSO4 F - MeOH G - Amchlor H - Ascorbic Acid I - Ice J - DI Water K - EDTA L - EDA Other: M - Hexane N - None O - AsNaO2 P - Na2O4S Q - Na2SO3 R - Na2S2O3 S - H2SO4 T - TSP Dodecahydrate U - Acetone V - MCAA W - pH 4-5 Z - other (specify)	
EFFLUENT DUPLICATE (190-25693-2)		Total Number of containers	
FIELD BLANK (190-25693-3)		Special Instructions/Note:	
 190-25693 Chain of Custody			
No. _____		This sample shipment is forwarded under chain-of-custody. If the laboratory does not currently maintain accreditation in the State of Origin listed above for analysis/tests/matrix being analyzed, the samples must be shipped back to the Eurofins TestAmerica laboratory or other instructions will be provided. Any changes to accreditation status should be brought to Eurofins TestAmerica attention immediately. If all requested accreditations are current to date, return the signed Chain of Custody attesting to said compliance to Eurofins TestAmerica.	
Possible Hazard Identification		Sample Disposal (A fee may be assessed if samples are retained longer than 1 month)	
Unconfirmed		☐ Return To Client ☐ Disposal By Lab ☐ Archive For _____ Months	
Deliverable Requested: I, II, III, IV, Other (specify)		Special Instructions/QC Requirements:	
Primary Deliverable Rank: 2		Method of Shipment:	
Empty Kit Relinquished by:		Date:	
Relinquished by: 		Date/Time: 4-15-21 1813	
Relinquished by:		Date/Time:	
Relinquished by:		Date/Time:	
Custody Seals Intact: A Yes A No		Custody Seal No.:	

ORIGIN ID: PHDA (330) 312-0176
EUROFINS TESTAMERICA CANTON
4101 SHUFFEL STREET NW
NORTH CANTON, OH 447208900
UNITED STATES US

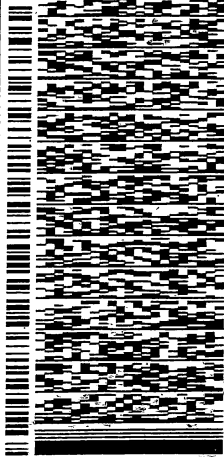
SHIP DATE: 15APR21
ACTWTG: 20.50 LB
CAD: 0562057/CAFE3409
BILL THIRD PARTY

TO ENVIRONMENTAL SAMPLE RECEIPT
TESTAMERICA BURLINGTON
30 COMMUNITY DRIVE

SOUTH BURLINGTON VT 05403

(802) 660-1800

DEPT: AL HAIDET



FedEx
Expt

FRI - 16 APR 10:3
PRIORITY OVERNIGHT

TRK# 9148 7503 7362
0801

NL BTVA

0541
VT-US B

